

SkyGuard AI | Use Cases & Validation Guide

Automatic Weather Station quality monitoring using temperature, pressure, and relative humidity. This guide accompanies the executable Python project.

1. Temperature spike

An unusual temporal score and abrupt change identify a possible spike or shift. The raw value is retained, alongside a provisional median for review.

2. Invalid humidity

Humidity above 100% raises a high-severity invalid-reading alert, even before the model has warmed up.

3. Missing pressure

A null, nonnumeric or nonfinite sample raises a missing-data alert. Investigate the sensor and transmission link.

4. Frozen pressure

Six identical samples trigger a frozen-sensor alert. Check firmware and sensor precision; quantization can produce legitimate repeats.

5. Communication loss

A gap above 15 minutes at the default 10-minute cadence is alerted when the next sample arrives. Immediate silence detection needs an upstream timer.

6. Regional weather event

Strong changes supported by two comparable peers for each changed variable can be marked as plausible weather. Production peer checks need time alignment and local suitability.

7. Degrading station health

At least 12 recent observations with 25% or more alerts mark the station for inspection. This is a maintenance cue, not a remaining-life prediction.

Evidence returned per observation

Each result preserves timestamp, station ID and all raw readings; alert, tentative fault type, severity, reasons, affected sensors, score, reference threshold, peer support, health and separate suggested values are added.

Reproducible demonstration

The shipped synthetic example detects 8 of 13 injected faulty readings and produces 10 false alerts among 851 normal readings: precision 0.4444, recall 0.6154, false-alert rate 0.01175. These numbers describe the example only.

Limits and next evaluation

Scores and evidence strength are not calibrated fault probabilities. This simulation cannot establish Indian AWS field performance. Reserve untouched stations and seasons, verify faults and real extreme weather, and report fault-wise recall, alert delay and false alarms per station-day. The Python package is not ESP32 firmware.